Semester 5 · Program Elective · 4 Credits · 60 periods

Material Handling

Handbook on selection, operation, economics, hoisting equipment, conveyors, surface transport, cranes, hoists, elevating equipment, safety and maintenance practices.

Course purpose

Material handling is the safe and economical movement, storage, control and protection of materials in a plant. It affects layout, productivity, safety, maintenance and cost.

Malayalam: Material handling □□□□□□□□□ □□□□□□□□ production delay, accident, product damage, maintenance cost □□□□□□□ □□□□□□.

Official details

Code: 5023B · **Title:** Material Handling · **Program:** Mechanical / Tool & Die / Manufacturing · **Prerequisite:** Industrial Engineering.

CO1

Classification, selection and economics.

CO2

Hoisting components, attachments and brakes.

CO3

Conveyors and surface transport.

CO4

Cranes, hoists and elevators.

Roadmap

CO1 · 11 Hours

Classification, Selection and Economics

Material handling system

Material handling equipment moves bulk load, unit load or discrete parts between workstations, stores and dispatch. Movements may be horizontal, vertical or combined. Good handling minimizes unnecessary movement and damage.

Unit load and palletization

Unit load combines smaller items into one handling unit. Palletization and containerization reduce handling time, counting effort and damage. Selection depends on load size, weight, route, distance, speed, layout and safety.

Maintenance and economics

Economics includes initial cost, running cost, labour saving, downtime, damage reduction, safety and flexibility. Maintenance includes inspection, lubrication, rope/chain check, brake check and alignment.

Expected questions

1. Classify material handling equipment.
2. Explain principles of material handling.

3. Explain palletization and containerization.
4. Explain maintenance points and economics.

CO2 · 14 Hours

Hoisting Appliances, Attachments and Brakes

Flexible hoisting appliances

Welded load chains, roller chains, hemp ropes and steel wire ropes are used for lifting. Selection depends on load, flexibility, fatigue, corrosion, factor of safety and fastening method.

Attachments

Hooks may be forged or triangular eye hooks. Load handling attachments include hook suspension appliances, crane grabs for unit or piece loads, electric lifting magnets and vacuum lifters.

Arresting gear and brakes

Arresting gear prevents uncontrolled movement. Brakes control starting, stopping and holding. Electromagnetic shoe brakes are common in hoisting and travelling mechanisms.

Expected questions

1. Explain wire rope and chain fastening methods.
2. Explain hooks, grabs, magnets and vacuum lifters.
3. Explain arresting gear and electromagnetic shoe brake.
4. Explain winches, capstans, turntables and monorail conveyors.

CO3 · 17 Hours

Conveyors and Surface Transport

Traction conveyors

Traction type conveyors use a moving traction element such as belt, roller, chain or bucket. Belt conveyors handle bulk material; roller conveyors handle packages; bucket elevators lift bulk material vertically. Escalators are also traction-type people/material movement systems.

Tractionless conveyors

Gravity conveyors use slope. Vibrating and oscillating conveyors move material by repeated motion. Screw conveyors push bulk material with rotating screw. Pneumatic and hydraulic conveyors use fluid flow.

Surface transport

Surface transport includes hand trucks, powered trucks, automatic guided vehicles and industrial trailers. AGVs follow programmed paths and support flexible manufacturing logistics.

Expected questions

1. Explain belt, roller and chain conveyors.
2. Explain bucket elevator and escalator construction.
3. Explain screw, pneumatic and hydraulic conveyors.
4. Explain AGV and powered trucks.

CO4 · 16 Hours

Hoists, Cranes and Elevating Equipment

Hoists

Hoists lift loads vertically. Lever operated, differential, worm geared and spur geared hoists differ in mechanism, speed, load capacity and effort required.

Cranes

Cranes move loads vertically and horizontally. Types include rotary, mobile, bridge, cable, floating, jib, derrick, overhead and gantry cranes. Selection depends on span, capacity, working area and duty cycle.

Elevating equipment and safety

Stackers, industrial lifts, passenger lifts and hydraulic lifts support vertical movement. Safety principles include rated load, limit switches, brakes, interlocks, inspection and trained operation.

Expected questions

1. Explain differential and worm geared hoists.
2. Compare overhead and gantry cranes.
3. Explain stackers, passenger lifts and hydraulic lifts.
4. List safety principles in material handling.

Rule bank

Minimize movement, use unit load, keep flow continuous, match equipment to load, inspect lifting devices, never exceed SWL and maintain brakes/ropes/chains.

Model paper

1. Explain classification and economics of material handling.

2. Explain hoisting chains, ropes, hooks and brakes.
3. Explain traction and tractionless conveyors.
4. Explain cranes, hoists, lifts and material handling safety.

Answer guide

For equipment answers write construction, working, application, advantages, limitations, maintenance and safety. For selection answers include load, distance, layout, cost, flexibility and risk.

References: Allegri, Apple, Rudenko, Alexandro, Oberman, K.C. Arora, R.B. Chaudhary and NPTEL/NDL resources.